

CountryEdu Assignment- Web Development

Assignment-2

Question 2: Regular Expression Matching

Given an input string P & Q , implement regular expression matching

+: One or more preceeding element

The matching should cover the entire input string.

Example 1:

Input: P = "AAAB" , Q="A+B"

Output: True

Explanation: '+' means one or more preceeding element 'A' , therefore by repeating 'AA' , it becomes AAAB

Example2:

Input: P = AABA Q=AA

Output: False

- P & Q contains only uppercase English Letters
- Pattern may contain invalid syntax – You must detect and return Invalid Pattern.

1.Approach

1. I will be using Two Pointer approach to solve this problem. I will be processing the string P and pattern Q from left to right.
2. The pointer i will indicate the current character in the string P and the pointer j will indicate the current character in pattern Q.
3. Firstly, I will check whether the pattern is valid or not. If + appears without a character before it, then the pattern is invalid.
4. If pattern contains + then it indicates that a character before + must appear one or more times in P.
5. After validating pattern I will proceed with comparing P and Q step by step. Both the string and pattern must be matched completely otherwise it will not be considered as a valid match.
6. I am using two pointers as for both P and Q there is a need to process them from left to right without repeatedly checking the characters that are already matched.

Example:

P = AAB

$$Q = A+B$$

In this case A+ matches AA and B matches B and hence the complete string is matched and result is True.

2. Algorithm

1. So we will Start with input string P and pattern Q.
2. First we will make sure Q is the valid pattern .
3. Invalid Pattern – if + occurs at the beginning or two + occur together in Q , then it is invalid pattern .
4. Initialize two pointers:
 - I = 0 for P
 - J = 0 for Q
5. We will traverse the pattern using pointer j .
6. If Q[j] is followed by + :
 - Before + character must occur at least once in P
 - While characters are equal keep moving i.
 - If in case no character is matched , return false .
 - Move j by 2 because both the character and + are processed.
7. If there is no + in Q[j] then normally compare P[i] and Q[j].
8. If they are equal , move both pointers forward.
9. If in any case they are not equal , return false .
10. As we process the pattern , we have to check whether the complete P has also been matched.
11. If both P and Q are processed completely , return true otherwise false .

3. C++ Code

```
#include <bits/stdc++.h>
using namespace std;

class Solution {
public:
    static bool solve(string &P, string &Q, int i, int j,
                     vector<vector<int>>> &dp) {

        if (j == Q.length()) {
            return i == P.length();
        }
        if (dp[i][j] != -1) {
            return dp[i][j];
        }
        bool hasPlus = (j + 1 < Q.length() && Q[j + 1] == '+');

        if (hasPlus) {
```

```

        if (i >= P.length() || P[i] != Q[j]) {
            return dp[i][j] = false;
        }

        return dp[i][j] = solve(P, Q, i + 1, j, dp) || solve(P, Q, i + 1, j + 2, dp);
    }
    if (i >= P.length() || P[i] != Q[j]) {
        return dp[i][j] = false;
    }
    return dp[i][j] =
        solve(P, Q, i + 1, j + 1, dp);
}

static string isMatch(string P, string Q) {
    for (int j = 0; j < Q.length(); j++) {

        if (Q[j] == '+' && j == 0) {
            return "Invalid Pattern";
        }
        if (Q[j] == '+' && Q[j - 1] == '+') {
            return "Invalid Pattern";
        }
    }
    int n = P.length();
    int m = Q.length();

    vector<vector<int>>> dp(
        n + 1,
        vector<int>(m + 1, -1)
    );

    return solve(P, Q, 0, 0, dp) ? "True" : "False";
}
};

```

4.Dry Run

Consider : P = "AAAB" , Q = "A+B"

So initially : i = 0 , j = 0

Here i points to first character of P and j points to the first character of Q

Step 1 : Check A+

Pattern starts with A+

In P, the first character is A, so we start matching A.

The first A matches, so

$i = 1$ - Now next character in P that is also A , so it matches .

$i = 2$ Again , character is A , so it matches as well.

$i = 3$ $P[i]$ is B , so A+ stops

Till now A+ has matched with AAA

Now we move j now $j = 2$

Step 2: Match B

Now $Q[j]$ is B.

Current character of P is also B.

So B matches with B .

So now : $i = 4$, $j = 3$ now $i == P.length()$ means complete String P has been matched with Q .So Result is True .

5.Output

Example 1 -

$P = \text{"AAAB"}$ $Q = \text{"A+B"}$ Output – True

Example 2 –

$P = \text{"AABA"}$ $Q = \text{"AA"}$ Output- False

Example 3 –

$P = \text{"AAAB"}$ $Q = \text{"+A"}$ Output – Invalid Pattern

6.Time and Space Complexity –

Time Complexity : $O(n+m)$

Where:

- n = length of string P
- m = length of pattern Q

Space Complexity : $O(1)$ because only few variables are being used .

